

King Fahd University of Petroleum and Minerals

College of Computing and Mathematics

Information and Computer Science Department

ICS 560: Fundamental of Quantum Computing

Semester 251

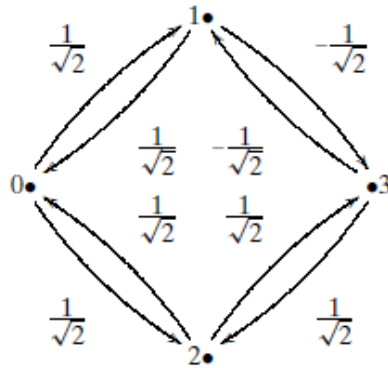
Assignment 2 (Probability Theory and Quantum Theory)

Show all your necessary steps:

1. How many bit strings of length 10 contain
 - a) exactly four 1s?
 - b) at most four 1s?
 - c) at least four 1s?
 - d) an equal number of 0s and 1s?
2. What is the probability that a fair die never comes up an even number when it is rolled **six** times?
3. A sequence of 10 bits is randomly generated. What is the probability that at least one of these bits is 1?
4. What is the probability that a positive integer not exceeding 100 selected at random is divisible by 5 or 7?
5. A pair of dice is loaded. The probability that a 4 appears on the first die is $2/7$, and the probability that a 3 appears on the second die is $2/7$. Other outcomes for each die appear with probability $1/7$. What is the probability of 7 appearing as the sum of the numbers when the two dice are rolled?
6. Let us assume that the particle is confined to $\{x_0, x_1, \dots, x_5\}$ and the current state vector is $|\psi\rangle = [2 - i, 2i, 1 - i, 1, -2i, 2]^T$. What is the probability of finding the particle at position x_1 ?
7. Do the vectors $|\psi\rangle = [1 + i, 2 - i]^T$ and $|\psi'\rangle = [2 + 2i, 1 - 2i]^T$ represent the same quantum state?
8. Normalize the ket $|\psi\rangle = [3 - i, 2 + 6i, 7 - 8i, 13i, 0]^T$.
9. Let the spinning electron's current state be $|\psi\rangle = 3i|\uparrow\rangle - 2|\downarrow\rangle$. Normalize the state and then find the probability that it will be detected in the up state.

10. Compute the amplitude of the transition from $|\psi\rangle = \frac{\sqrt{2}}{2} [i, -1]^T$ to $|\phi\rangle = \frac{\sqrt{2}}{2} [1, -i]^T$

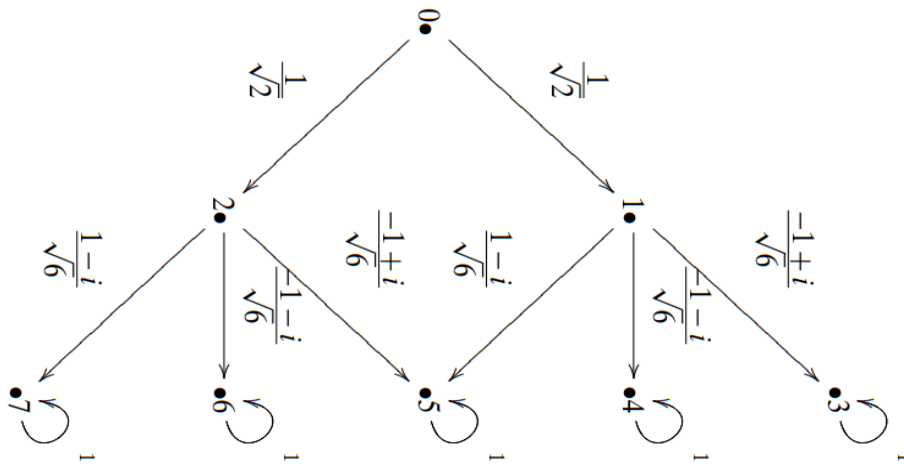
11. Consider the following graph



a) Construct the adjacency matrix of the above graph.

b) Is the resulting matrix unitary?

12. Consider the quantum system given by the following graph.



a) Construct the adjacency matrix of the system.

b) If the particle at state $X = \left[\frac{1}{\sqrt{3}}, \sqrt{\frac{1}{5}}, 0, \frac{i}{\sqrt{15}}, 0, \frac{i}{\sqrt{15}}, \sqrt{\frac{2}{15}}, \sqrt{\frac{1}{5}} \right]^T$. Find the state of X after one movement